

SWYNEX Internship — Final Data Analytics Project

Superstore Sales & Profit Analysis

Tools: Python, Pandas, NumPy, Matplotlib, Power BI and GitHub

1. Problem Statement

Analyze a retail Superstore dataset to assess sales, profitability, customer segments, regional performance, discount patterns and product-level performance. Clean the data, perform exploratory analysis, identify actionable business insights, and communicate the findings through an interactive Power BI dashboard.

2. Dataset Information

Item	Details
Dataset	Sample Superstore
Original records	9,994
Final records	9,977
Columns	13
Measures	Sales, Quantity, Discount, Profit
Dimensions	Category, Sub-Category, Region, Segment, Ship Mode, State, City

3. Data Cleaning Process

- Inspected all 13 columns; no missing values were found.
- Removed 17 exact duplicate records.
- Validated numerical and categorical data types.
- Checked categorical values for inconsistent spelling/capitalization and whitespace.
- Validated Sales, Quantity and Discount ranges.
- Retained negative Profit values because they represent legitimate business losses.
- Final validation: 9,977 rows, 13 columns, 0 missing values and 0 duplicate records.

4. Exploratory Analysis

Metric	Result
Total Sales	\$2,296,194.59
Total Profit	\$286,238.45
Average Sales	\$230.15
Median Sales	\$54.82
Average Profit	\$28.69
Median Profit	\$8.67
Total Quantity Sold	37,820
Overall Profit Margin	12.47%

5. Key Business Insights

- Technology generated the highest category sales and profit.
- West was the strongest region in both sales and profit.
- Consumer was the largest customer segment by sales, profit and quantity.
- Tables was the largest loss-making sub-category despite substantial sales.
- Bookcases and Supplies also recorded negative total profit.
- Copiers generated the highest sub-category profit despite relatively low quantity sold.
- Discount and profit showed a weak negative correlation of -0.22; this indicates association, not causation.
- Discount levels from 30% to 80% recorded negative total profit in this dataset.
- The highest-sales transaction was not necessarily profitable.

6. Interactive Power BI Dashboard

The dashboard combines KPI cards, category analysis, regional analysis, customer-segment analysis, sub-category profitability and discount-level profitability. Interactive slicers allow filtering by Region, Category, Segment and Ship Mode.

7. Project Deliverables

- data/cleaned_superstore.csv
- notebooks/data_cleaning.ipynb
- notebooks/SWYNEX_Task2_Exploratory_Analysis.ipynb
- dashboard/SWYNEX_Task3_Dashboard.pbix
- dashboard/SWYNEX_Task3_Dashboard.png
- SWYNEX_Final_Project_Report.pdf

8. Conclusion

This project demonstrates an end-to-end data analytics workflow: data quality assessment and cleaning, exploratory analysis, business insight generation and interactive dashboard development. The findings highlight where sales and profits are concentrated and identify product and discount areas that require closer business attention.